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The Effects of Force-Velocity Profile-Based Load Prescription on Muscle Power and Functional Performance in Velocity-Deficient Older Adults

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ABSTRACT

Muscle power is essential for functional capacity in older adults and can be improved through power-focused resistance training (PRT). PRT adaptations are load- and velocity-specific, yet optimal load prescription remains unclear. As force and velocity capacities vary between individuals, this 12-week intervention compared two force-velocity (F-V) profile-guided loading strategies: training at loads matched versus not matched to the F-V deficit. F-V profiles of 80 adults ≥ 65 years were assessed on a pneumatic leg press, with maximal force (F_0) and velocity (V_0) expressed as z-scores relative to a younger reference group ($n = 60$, 20–35 years). A difference ≥ 1 between z-scores was considered indicative of a deficit, with the direction determining a force- or velocity-deficit profile. As force deficits were rare ($n = 4$), the intervention only included participants with a velocity deficit ($n = 59$), of whom 57 initiated the intervention. Participants were randomly assigned to train at loads matched to their deficit (individualized; IND) or at loads deliberately not matched to their deficit (non-individualized; NON-IND). Maximal power, F-V profile parameters and functional performance were assessed at baseline, 6 and 12 weeks. From baseline to 12 weeks, F_0 increased more in NON-IND than IND ($p = 0.040$), while V_0 increased more in IND than NON-IND ($p = 0.009$). Changes in power and functional performance did not differ between groups (all $p \geq 0.065$). In velocity-deficient older adults, differences in F-V-guided load prescription led to distinct F-V adaptations but did not lead to differences in power or functional performance.

Trial Registration: The study was registered at ClinicalTrials.gov (NCT06449716)

1 | Introduction

Muscle power declines progressively with age [1, 2], reducing physical function and increasing the risk of falls, disability, and mortality [3]. Power-focused resistance training is a safe and effective way to counter these declines [4]. In older adults, power training has been studied across a broad range of loading intensities, with benefits reported at low, moderate, and high loads [3, 5]. As improvements in muscle power have been observed across this loading spectrum [6, 7], no single optimal training intensity has been established.

Although overall improvements in muscle power may appear similar across loading schemes, the underlying adaptations can differ. Muscle power is determined by both force and velocity, which are inversely related. As loading intensity and consequently muscle force demands increase, movement velocity decreases [8]. This relationship can be described using

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the force-velocity (F-V) profile, which spans from maximal force at zero velocity (force capacity) to maximal velocity at zero force (velocity capacity) [9]. Evidence suggests that low-load power training induces greater improvements in velocity-related capacities, whereas high-load power training leads to larger gains in force-related capacities, thereby altering the overall F-V profile [7]. Notably, considerable inter-individual variability in training responses has been reported. In a previous 12-week intervention comparing low-load and high-load power training [7], the standard deviation of changes in maximal power exceeded the mean improvement in both groups. This large between-individual variability suggests that responses to a given power training protocol may differ substantially between individuals.

A potential explanation for this variability is that age-related declines in muscle power may arise from deficits in either force or velocity, and the relative contribution of each differs between individuals [1, 9]. Because training adaptations are load- and velocity-specific [7], assigning a person with a predominant velocity deficit to a force-focused program (or vice versa) may limit their potential gains. Considering individual F-V profiles when prescribing training loads could therefore help optimize training adaptations in older adults. Evidence from young athletes supports this personalized approach: tailoring training to each participant's F-V profile produces favorable shifts in the F-V relationship and greater performance gains than generic programs [10]. However, other evidence suggests that shifting the entire F-V curve rightward and improving power across the full continuum, regardless of initial F-V profile, may matter more than correcting a theoretical F-V imbalance [11, 12].

In older adults, evidence is limited. One study compared individualized F-V-based training, where participants trained according to their deficit, to a generic program spanning a wide range of the F-V spectrum [13]. Although individualized training did not demonstrate greater improvements in muscle power and functional performance, several factors provide important context for interpreting the findings of this study [13]. First, participants were categorized into force and velocity deficit groups using a median split of the F-V slopes within the sample. Consequently, individuals with intermediate values may have been assigned to different deficit groups despite exhibiting similar profiles. The authors noted that tertile-based analyses produced comparable results; however, additional details on how the middle tertile was handled and to what extent this impacted sample size or statistical power would help contextualize this observation. Moreover, the median-split categorization assumes that force and velocity deficits occur with equal frequency, which has not been verified and may not reflect the true distribution of deficits in the sample. Second, participants in the individualized condition trained within one portion of the F-V spectrum, whereas those in the generic group trained across a broad range. As a result, the difference between the training stimuli between groups was relatively subtle and may explain similar responses to training. Finally, the force- and velocity-targeted subgroups were combined into a single “individualized” condition. Functional outcomes that depend more strongly on either force or velocity might respond differently to targeted improvements in these components, and combining the subgroups may have made such distinctions less visible.

Building on that work [13], the present study used an alternative methodological approach to study the effects of F-V profile-based loading strategies in older adults. To address the assumption that force and velocity deficits are equally distributed among older adults, participants were classified into deficit groups based on how they compared to a young reference group rather than their position within the sample. To avoid overlap between training conditions, we compared training with loads matched to an individual's predominant F-V deficit (individualized; IND) with training at loads deliberately not matched to that deficit (non-individualized; NON-IND), rather than with a mixed-load approach. We examined whether IND training improves muscle power and functional performance more than NON-IND training.

We hypothesized that older adults who train with loads matched to their predominant F-V deficit would show greater improvements in muscle power and functional performance than those who train with loads deliberately not matched to that deficit. We further expected these gains to be accompanied by greater adaptations in the F-V component identified as the predominant deficit at baseline.

2 | Methods

2.1 | Study Design

The study was a randomized controlled trial investigating two F-V profile-guided training approaches. Older adults exhibiting a clear deficit in either force or velocity relative to a young reference group were eligible for inclusion. Participants that were included in the intervention were randomly assigned to one of two groups: a group training with loads matched to their individual F-V deficit (IND), or a group training with loads deliberately not matched to that deficit (NON-IND). Muscle power, F-V profile parameters and functional performance were assessed at baseline (pre-intervention; PRE), 6 weeks (mid-intervention; MID) and 12 weeks (post-intervention; POST).

The intervention was conducted at KU Leuven, approved by the Ethical Committee UZ/KU Leuven (S68434) and registered at ClinicalTrials.gov (NCT06449716). All participants provided written informed consent in accordance with the Declaration of Helsinki. Generative artificial intelligence (AI) tools were used during the conduct of this study. Claude (Anthropic; Sonnet-series model) was used to assist with the development of statistical analysis code in R. Claude (Anthropic; Sonnet-series model), ChatGPT (OpenAI; GPT-5-based model), and Microsoft 365 Copilot (Microsoft; GPT-5-based model) were also used to assist with language editing and refinement of the manuscript text. All AI-assisted outputs were critically reviewed and verified by the authors, who retain full responsibility for the accuracy, integrity, and scientific content of the manuscript, analyses, and conclusions.

2.2 | Participants

A group of healthy young adults (20–35 years; $n = 60$; 30 males, 30 females) was recruited to establish reference F-V profiles. A separate group of community-dwelling older adults (≥ 65 years; $n = 80$; 50 males, 30 females) was recruited for baseline testing. Recruitment was conducted through newspaper advertisements, information sessions at leisure organizations, flyers, and word-of-mouth.

2.2.1 | Eligibility Criteria

For both the young and older groups, exclusion criteria included unstable cardiovascular disease, neurological disorders, inability to follow instructions, low physical function (Short Physical Performance Battery score < 7), acute infection or fever, and severe musculoskeletal conditions. In addition, participants were required to have no regular engagement in resistance training (≥ 1 session/week) during the previous 12 months and no participation in competitive sports. Participation in recreational endurance exercise and other physical activities was permitted.

In the older group, not all participants measured at baseline were retained for the intervention. Inclusion depended on how each participant's F-V profile compared with the young sex-specific reference group. Z-scores were calculated for estimated maximal isometric force (F_0) relative to body mass (BM) and estimated maximal velocity (V_0):

$$z = \frac{X_{\text{old}} - \mu_{\text{young}}}{\text{SD}_{\text{young}}}$$

The magnitude and direction of the difference between the z-scores for F_0 and V_0 determined the participant's predominant deficit type. Only participants with an absolute difference of at least one unit between z-scores (i.e., $|zF_0 - zV_0| \geq 1$) were classified as having a deficit and included in the intervention. Participants with a difference < 1 were classified as neutral and excluded. This threshold was selected to ensure meaningful differences between z-score-based force- and velocity-deficit groups. As it represents an operational classification rule rather than a validated physiological cut-point, the terms 'force deficit' and 'velocity deficit' should be interpreted as study-specific classifications rather than physiological diagnoses.

2.2.2 | Sample Size

Figure 1 presents the planned classification of participants into force-deficit and velocity-deficit groups and their allocation to IND or NON-IND training. Separate a priori power analyses were performed within each deficit group. The analysis (G*Power 3.1, repeated-measures ANOVA, within-between interaction) assumed a medium effect size (partial $\eta^2 = 0.06$), $\alpha = 0.05$, power = 0.95, two groups (IND and NON-IND), three measurement occasions (PRE, MID, and POST), a correlation of $r = 0.75$ among repeated measures, and a non-sphericity correction of $\varepsilon = 0.75$. Results indicated that 28 participants were required per deficit group. Allowing for an anticipated 10% dropout rate, the planned sample size was increased to 32 participants per deficit group (32 force deficit and 32 velocity deficit), with each group allocated to IND or NON-IND training ($n = 16$ per training group), resulting in a total target sample of 64 participants.

Figure 2 presents participant inclusion and retention throughout the intervention. Of the 153 older adults who expressed interest and were informed about the study set-up and requirements, 80 met the eligibility criteria and completed the baseline assessment. Of these, 17 (21.25%) had a neutral profile, 4 (5%) had a force deficit, and 59 (73.75%) had a velocity deficit.

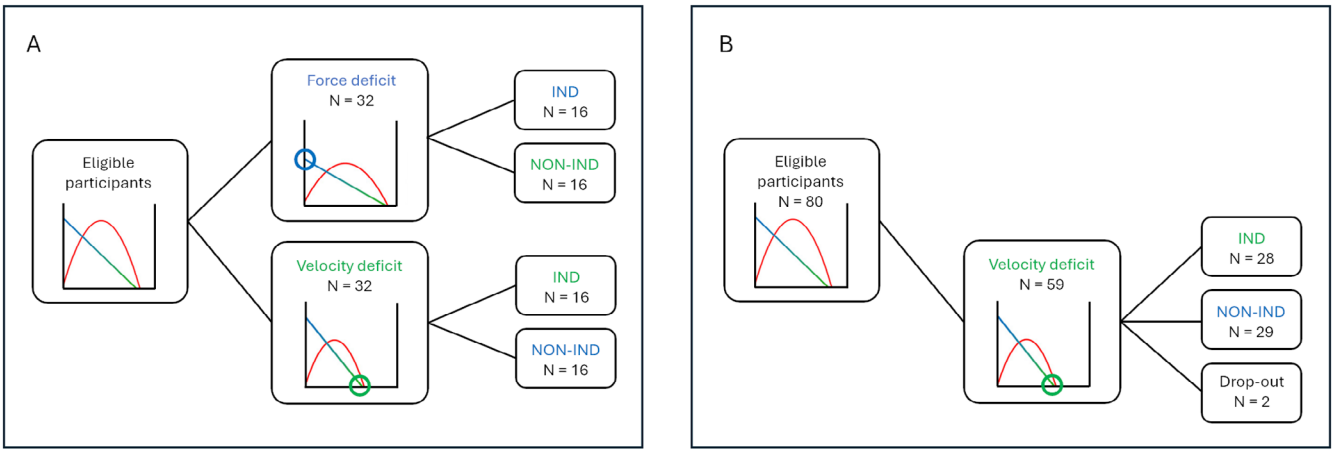


FIGURE 1 | Pre-specified (A) versus realized (B) participant classification and allocation. Force (blue), velocity (green) and power (red) relationships are shown, with circles indicating the identified deficit type. Panel A illustrates the planned classification and allocation based on the a priori power analysis, which estimated 32 participants with a force deficit and 32 with a velocity deficit, each randomized to an individualized (IND) or non-individualized (NON-IND) training group ($n = 16$ per group). Panel B presents the realized allocation. Of the 80 participants assessed, 59 were classified as having a velocity deficit, 4 as having a force deficit and 17 as having a neutral profile. Owing to the low prevalence of force deficit, this intervention arm was discontinued. Among participants with a velocity deficit, two withdrew before the start of the intervention, 28 were allocated to the IND group and 29 to the NON-IND group. The intervention and subsequent analyses were restricted to participants classified as velocity-deficient.

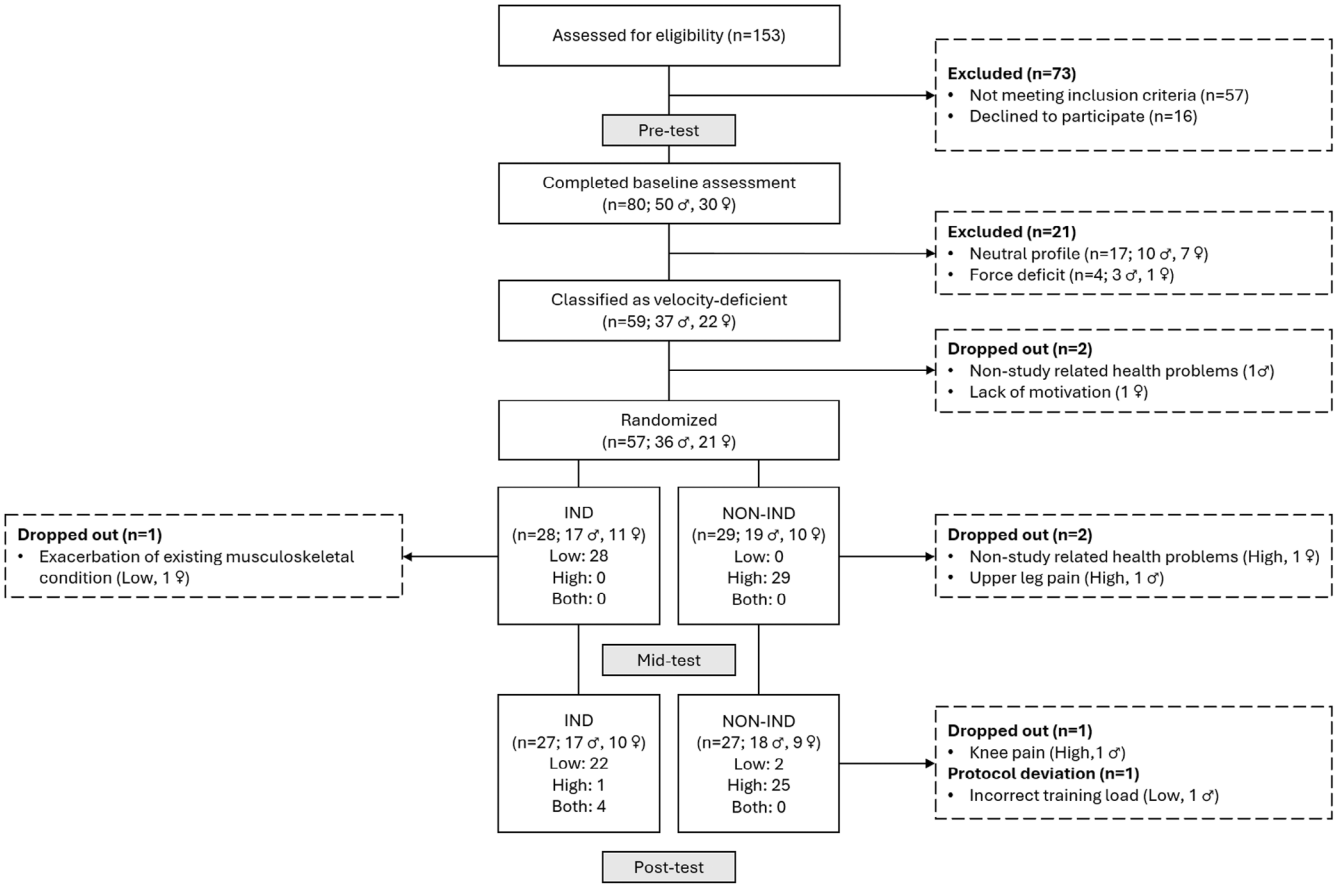


FIGURE 2 | Participant flow through the study. The flowchart presents the number of participants screened for eligibility, excluded, randomized and allocated to individualized (IND) or non-individualized (NON-IND) training, including discontinuations and their reasons. Of the 57 participants who started the intervention, 54 were included in the MID analyses (IND: $N = 27$; NON-IND: $N = 27$) and 52 were included in the POST analyses (IND: $N = 27$; NON-IND: $N = 25$). The distribution of training received in these groups is indicated in the figure: Low load, high load or both (one session low and one session high each week). These sample sizes apply to all leg press outcomes and most functional performance outcomes. Exceptions were stair-climb outcomes, for which 54, 53, and 48 observations were included at PRE, MID and POST, respectively (due to processing errors) and the countermovement jump, for which one participant did not perform the test at any time point.

The distribution of deficit profiles was similar between males and females. Among males, 20% had a neutral profile, 6% a force deficit, and 74% a velocity deficit. Among females, the corresponding proportions were 23.33%, 3.33%, and 73.33%, respectively.

The low prevalence of force deficits made it unfeasible to achieve the planned sample size for this subgroup. Recruitment was initially continued beyond the target sample size in an attempt to identify additional participants with a force deficit. However, after 80 participants had been assessed, recruitment was discontinued and subsequent analyses were restricted to participants with a velocity deficit. As a result of the extended recruitment period, the number of participants with a velocity deficit ($n = 59$) substantially exceeded the planned sample size ($n = 32$). Among the 59 participants with a velocity deficit, two withdrew at the start of the intervention. The remaining 57 participants (36 males, 21 females) were randomly allocated (1:1) to the IND or NON-IND training group using a computer-generated randomization sequence stratified by age and sex.

To evaluate the test–retest reliability of the deficit classification procedure, a subset of 32 participants (17 males, 15 females) completed two test sessions separated by 1 day to 1 week. Twenty-eight of 32 participants (87.5%) received the same classification at both assessments. Cohen's kappa was 0.65, indicating substantial agreement. All four participants (12.5%) who changed classification switched between the neutral and velocity-deficit categories (two in each direction). No participants switched between the force- and velocity-deficit categories. Although the classification procedure demonstrated substantial test–retest agreement, participants with values close to the classification threshold may change category across repeated assessments due to measurement error and day-to-day variability.

2.3 | Training Intervention

Figure 3 provides an overview of the training progression in the IND and NON-IND training programs. The 12-week intervention consisted of twice-weekly sessions performed on non-consecutive days. All sessions were conducted in a

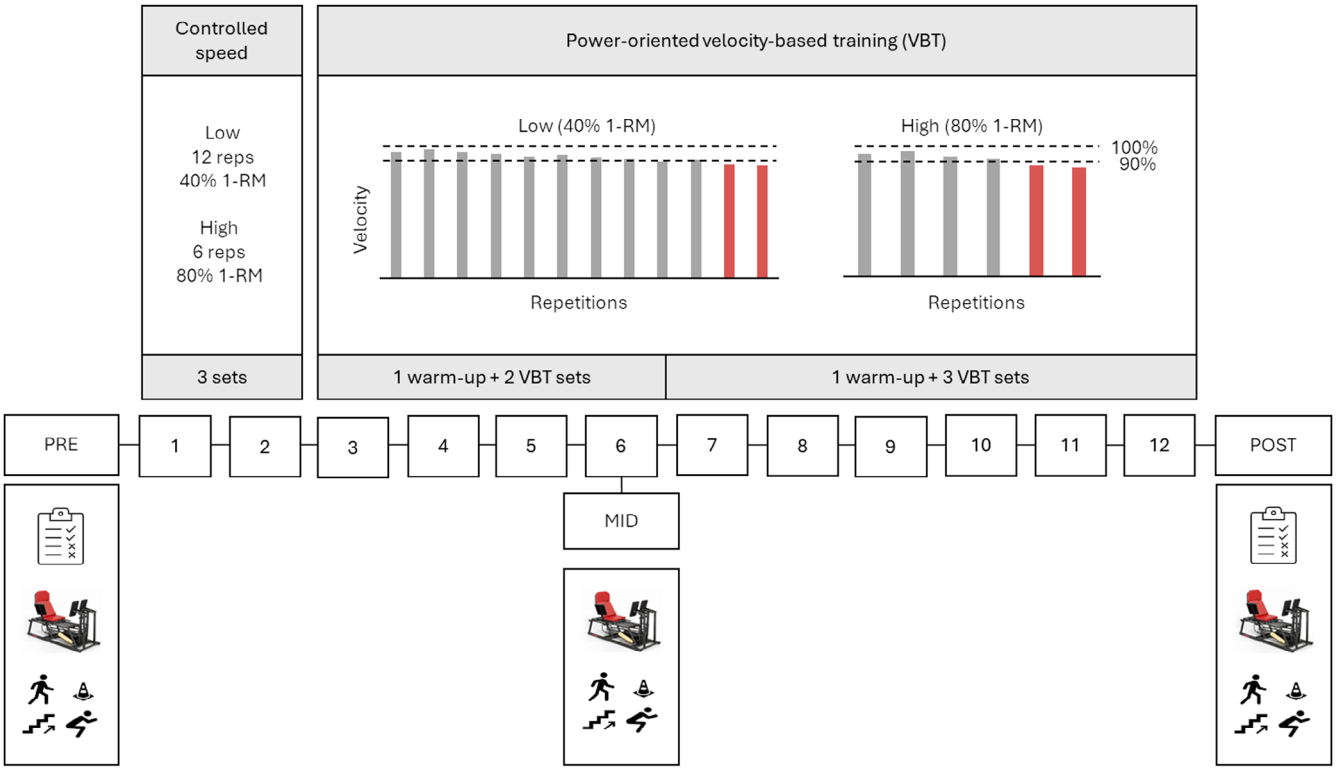


FIGURE 3 | Overview of the 12-week, power-focused training intervention. Participants completed supervised unilateral leg press sessions twice weekly on non-consecutive days. Training loads were determined by intervention group (IND vs. NON-IND). Low load was set at 40% and high load at 80% of one-repetition maximum (1-RM). Sessions progressed from controlled-speed to power-oriented sets. Following a velocity-based training (VBT) method, power-oriented sets were terminated using a 10% within-set velocity loss criterion. Load, repetitions, and movement velocities were monitored across the intervention. Leg press force-velocity profile parameters and functional performance were assessed pre-, mid-, and post-intervention. Pre-intervention questionnaires captured demographics, health status, and physical activity levels. Post-intervention questionnaires captured changes in diet, health status, and physical activity, as well as satisfaction with training group allocation.

laboratory-based gym setting and supervised by appropriately trained graduates in sport sciences or physiotherapy. Each session began with a 10 min warm-up on a cycle ergometer (Technogym Bike Excite). Participants then trained both legs unilaterally on a pneumatic leg press device (Keiser Sport, Fresno, USA). Dropout rates, adherence, adverse events and training parameters (i.e., load, repetitions, and the maximal and minimal velocities within each set) were recorded throughout the intervention. Leg press F-V profile parameters and functional performance were assessed pre-, mid- and post-intervention. Participants were instructed to maintain their usual physical activity levels and dietary habits throughout the study.

Questionnaires were completed pre- and post-intervention. Physical activity and sedentary behavior were assessed at both time points using the International Physical Activity Questionnaire Short Form (IPAQ-SF) [14]. Pre-intervention questionnaires additionally collected demographic information and information on health status. Post-intervention questionnaires asked participants whether they had experienced changes in health status or diet during the intervention (yes/no). Those responding affirmatively were asked to provide additional details to identify changes that could potentially influence study outcomes. These data were collected for monitoring purposes only and were not included as covariates in the statistical analyses because they were not designed to provide quantitative measures. Participant satisfaction with group allocation was assessed by asking whether they would choose low-load or high-load power training if given the option.

2.3.1 | Load Assignment and Adjustments

As described in Sections 2.2.1 (Eligibility Criteria) and 2.2.2 (Sample Size), all randomized participants who started the intervention had a velocity deficit at baseline. The training load in the first 6 weeks was therefore low (40% of one-repetition maximum; 40% 1-RM) for IND and high (80% 1-RM) for NON-IND. After the mid-intervention assessment, pre-specified load adjustments were made whenever continuing the baseline program would no longer have corresponded to a participant's allocation. In IND, participants with a neutral profile ($n = 4$) trained at both low and high loads (one session per week at each), because continuing low-load training would no longer have matched a deficit. The participant with a force deficit ($n = 1$) trained exclusively at high loads, matching the new deficit. In the NON-IND group, participants with a neutral profile ($n = 2$) continued training at high loads, because switching to mixed- or low-load training would have aligned more with their updated F-V profile. The participant with a force deficit ($n = 1$) switched to low-load training, because continuing high-load training would have matched the new force deficit. In both groups, participants that showed no

change in deficit ($n = 46$) continued to train according to their original load prescription.

2.3.2 | Training Progression and Monitoring

During the first 2 weeks, participants performed a fixed volume of three sets of 12 repetitions at 40% 1-RM or six repetitions at 80% 1-RM in the IND and NON-IND group, respectively. Movement speed was controlled at 1 s for the concentric phase and 3 s for the eccentric phase. From week 3 onward, sessions began with a preparatory set at controlled speed, followed by power-oriented sets (two sets in weeks 3–6, three sets in weeks 7–12) with the concentric phase as fast as possible and the eccentric phase at controlled speed (3 s). Repetitions were prescribed using a within-set velocity loss of 10%. More specifically, sets ended when concentric velocity dropped > 10% from the fastest repetition for two consecutive repetitions. This threshold was selected based on evidence that it improves lower-limb strength and movement velocity in older adults while minimizing training volume [15]. To provide a more comprehensive exercise program, upper-limb exercises (chest press and low row; 2 × 12–15 repetitions to volitional fatigue at a controlled tempo) were added. Outcome measures, however, were limited to lower-limb function.

The training data was recorded throughout the intervention period. Maximal velocity (V_{max}; $m \cdot s^{-1}$) was defined as the velocity of the fastest repetition within a set, whereas minimal velocity (V_{min}; $m \cdot s^{-1}$) was defined as the slower of the two consecutive repetitions that exceeded the prescribed velocity loss threshold within a set. Within-set velocity loss was calculated as $([V_{max} - V_{min}]/V_{max}) \times 100$. Set volume index was calculated for each completed set as repetitions × relative load (%1-RM), and mean set volume per participant was obtained by averaging these values across all completed sets. Session volume index was calculated as the sum of set volumes within a session, and mean session volume per participant was obtained by averaging session volumes across all completed sessions. Volume indices were used as practical estimates of training volume and should not be interpreted as measures of mechanical work or overall training stimulus. Training variables were compared between groups and within groups using independent and paired *t*-tests, respectively.

2.4 | Outcome Measures

2.4.1 | Muscle Power and F-V Profiling

Muscle power testing and F-V profiling were conducted on a Keiser A400 horizontal pneumatic leg press (Keiser Sport, Fresno, USA) based on a validated procedure for older adults [16]. Testing was performed on the dominant leg only to reduce participant burden and testing duration. This approach is supported by previous research in older adults reporting minimal inter-limb asymmetries across a broad range of lower-limb muscle function outcomes, with asymmetries observed only for single-joint isotonic peak velocity [17]. Furthermore, F-V profile parameters are derived from force and velocity measurements obtained across multiple loading conditions rather than from a single performance outcome, making them less susceptible to load- or velocity-specific inter-limb differences.

Seat position was individually adjusted to ensure a 90° knee angle at the start of the concentric phase [18]. The non-dominant leg rested on the contralateral footplate stop and participants held the seat handles for stability. Each testing session began with a standardized warm-up of 5 min of self-paced cycling (Technogym Bike Excite), followed by two specific warm-up sets on the test device: 10 repetitions at 15% and 6 repetitions at 30% of body mass. The warm-up repetitions were performed at a controlled tempo, except for the final 2–3 repetitions of the 6-repetition set, which were performed with maximal intended movement velocity.

2.4.1.1 | Familiarization. At least 48 h before baseline testing, participants completed a familiarization session to learn proper positioning, movement technique, and breathing strategy (to avoid the Valsalva maneuver). During familiarization, participants performed single, explosive concentric leg extensions at six submaximal loads scaled to body mass (7.5%, 15%, 30%, 45%, 60%, and 75%). They completed three trials at the three lighter loads and two trials at the three heavier loads. The fastest repetition at each load was used for analysis. Using the procedures described in the *Data Processing and Outcomes* section, these six load-velocity pairs were used to generate a preliminary F-V profile and to estimate F_0. If the last load (75% of body mass) corresponded to less than 70% of the estimated F_0, an additional load equal to 70% of the estimated F_0 was added to the protocol.

2.4.1.2 | Testing Procedure. At each testing time point (pre-, mid- and post-intervention), participants started by performing the same protocol as during the familiarization. However, the five progressive loads were prescribed relative to their estimated F_0 from the familiarization session (12.5%, 25%, 37.5%, 50% and 62.5% of estimated F_0) rather than body mass. The same absolute loads were used at the pre-, mid- and post-intervention tests to assess participants under comparable external loading conditions over time. Consequently, mid- and post-test loads were not adjusted based on updated F_0 estimates from the preceding test. The three lighter loads were performed three times each, and the two heavier loads twice, with the repetition showing the highest mean concentric velocity retained for analysis. Following these trials, participants completed single repetitions at incremental loads to determine 1-RM. Rest intervals were adjusted according to the load of the preceding trial: 60 s for the three lightest loads, 90 s for the two heavier loads and 120 s for the additional repetitions up to 1-RM.

2.4.1.3 | Data Processing and Outcomes. Load was used as a proxy for force, and mean concentric velocity was measured using the device's integrated software, which calculates piston displacement and movement velocity [18]. During testing, force and velocity data from each repetition were entered into a Microsoft Excel template that continuously updated the participant's F-V relationship [16]. The F-V relationship was assumed to be linear [9]. During testing, data points that deviated from linearity were re-evaluated. Specifically, when the highest mean velocity achieved against a given load differed by more than 0.03 $m \cdot s^{-1}$ from the value predicted based on the preceding loads, an additional repetition was performed at that load.

If the deviation remained greater than 0.03 m·s^{-1}, the load was excluded and the final F-V relationship was calculated using the remaining loads. The regression equation was subsequently used to derive the maximum theoretical force (F_0; N), maximum theoretical velocity (V_0; m·s^{-1}), F-V slope ($-F_0/V_0$; N·s·m^{-1}), and maximal power output ($P_{max}=[F_0 \times V_0]/4$; W). F_0 and P_{max} were additionally normalized to body mass.

On average, 6.08 ± 1.41 points were retained and 2.58 ± 1.27 points were discarded per assessment and participant, with the resulting regressions showing excellent linearity ($R^2 > 0.94$). The numbers of retained and discarded observations were similar across groups and time points, ranging from 5.78 to 6.40 and from 2.19 to 2.93, respectively.

Test–retest reliability was evaluated in 32 participants (17 males and 15 females), each of whom completed two test sessions separated by 1 day to 1 week. For the F-V variables, intraclass correlation coefficients (ICC; two-way mixed-effects model, absolute agreement, single measures) were 0.99 (95% CI: 0.97–1.00) for P_{max} (W·kg^{-1}), 0.98 (95% CI: 0.96–0.99) for F_0 (N·kg^{-1}), and 0.94 (95% CI: 0.87–0.97) for V_0 (m·s^{-1}), indicating excellent reliability. Corresponding within-subject coefficients of variation were 2.6% ± 3.0%, 2.9% ± 2.4%, and 2.2% ± 1.6%, respectively. The standard errors of measurement (SEM) were 0.16 W·kg^{-1} (3.06%) for P_{max}, 0.44 N·kg^{-1} (3.69%) for F_0, and 0.04 m·s^{-1} (2.61%) for V_0. The z-score difference, which served as the classification measure, also demonstrated excellent reliability, with an ICC of 0.928 (95% CI: 0.858–0.964) and a SEM of 0.468 z-score units.

2.4.2 | Functional Performance

Functional performance was assessed using a test battery administered in the following order: handgrip strength (HGS), 10 m maximal gait speed (MGS), five-repetition sit-to-stand (5×STS), timed up-and-go (TUG), countermovement jump (CMJ), stair ascent (SA), and 30 s sit-to-stand (30sSTS). During the SA and STS tests, an inertial measurement unit (DynaPort MoveTest, McRoberts, The Hague, The Netherlands) was placed on the lower back. Data were recorded at 100 Hz and analyzed using the manufacturer's software.

2.4.2.1 | HGS. Maximal grip strength (kg) of the dominant hand was measured using a Jamar dynamometer adjusted for hand size. Participants were seated with a straight back, feet flat, and the non-dominant arm resting on their legs. The dominant elbow was flexed at 90° with the forearm in a horizontal, neutral position. They squeezed the dynamometer maximally for 5 s. Three trials were performed, and the highest value was retained for analysis.

2.4.2.2 | MGS. Participants walked 10 m as fast as possible without running. Timing gates (Racetime2 Light Radio, Microgate, Italy) recorded the time (s). Three trials were performed and the fastest time was retained for analysis.

2.4.2.3 | TUG. Participants began seated with their back against a standardized chair (0.46 m height, no armrests). They were instructed to stand, walk 3 m to a cone, turn 180°, return and sit back with their back touching the chair. The task was performed as fast as possible, with time (s) recorded using a stopwatch. Three trials were conducted, and the fastest time was used for analysis.

2.4.2.4 | CMJ. Participants performed a vertical countermovement jump with their hands on their waist. Flight time (t, s) was measured using a custom-built contact mat and converted to jump height (h, m) using the equation $h = gt^2/8$, where g is the acceleration due to gravity (9.81 m·s^{-2}) [19]. Three trials were performed, and the trial with the highest jump height was retained for analysis.

2.4.2.5 | SA. Participants ascended a six-step staircase (18 cm rise, 26.5 cm going, 4.5 cm nosing) as fast as possible without skipping steps or using the handrail. Three trials were performed. The trial with the fastest ascent duration (s) based on sensor data was used for analysis. For this trial, the mean power during the rise phase of each step was calculated following established procedures [20], and the average power across all six steps (W·kg^{-1}) was used as the outcome measure.

2.4.2.6 | STS. The 5×STS and 30sSTS were performed on a standardized chair (0.46 m height, 0.42 m depth, no armrests). Participants completed sit-to-stand cycles as quickly as possible, with arms crossed over the chest and back touching the chair at the start and end of each repetition. For the 5×STS, total duration (s) was calculated from the start of the movement until the fifth repetition ended in a standing position [21]. Three trials were performed, and the fastest trial based on sensor data was used for analysis. For the 30sSTS, participants performed a single trial, and the total number of repetitions completed in 30 s was recorded. In both STS tests, mean power during the sit-to-stand phase of each repetition was calculated [20], and the average power across all cycles (W·kg^{-1}) was used as the outcome measure.

2.5 | Statistical Analysis

All analyses were conducted in R (version 4.4.2), with statistical significance set at $\alpha = 0.05$. Changes in outcome variables across three time points (PRE, MID, POST) were analyzed using linear mixed-effects models fitted via maximum likelihood (ML); final models were refitted using restricted maximum likelihood (REML) for parameter estimation. Participant ID was included as a random intercept to account for interindividual baseline differences. Two model specifications were applied to each outcome. A time-only model (outcome~time point + (1|ID)) assessed overall changes across time points irrespective of group allocation. A Time × Group interaction model (outcome~time point × group + (1|ID)) evaluated whether temporal changes differed between the IND and NON-IND groups.

Estimated marginal means were obtained using the *emmeans* package. Three pairwise contrasts (PRE–MID, MID–POST, PRE–POST) were constructed. For the time-only model, contrasts were calculated across the full sample, whereas for the interaction model, contrasts were computed separately within each group. Bonferroni correction was applied to adjust for

multiple comparisons. Within-group effect sizes were calculated as Hedges' g_{av}, using the mean within-individual change score standardized by the average of the pre- and post-assessment standard deviations [22]. Effect sizes were interpreted as negligible (<0.2), small (0.2–0.5), medium (0.5–0.8) or large (>0.8).

Model assumptions were evaluated through visual inspection of diagnostic plots. To assess robustness, an outlier detection procedure was implemented. Studentized residuals were computed for each fitted mixed-effects model. Observations exceeding the Bonferroni-corrected critical t value ($\alpha = 0.05$) were flagged as potential outliers. Results from the primary and sensitivity analyses (excluding outliers) were compared to determine if statistical conclusions were affected.

3 | Results

3.1 | Sample Characteristics

Table 1 presents the descriptive characteristics and F-V profile parameters of the young reference group ($n = 60$) and the sample of older adults assessed at baseline ($n = 80$).

In the older group, all participants scored ≥8 on the Short Physical Performance Battery (SPPB), and most achieved the maximum score (median: 12, IQR: 11–12), indicating a high-functioning sample.

Physical activity and sedentary behavior were assessed in both groups. In the young group, the median frequency of vigorous, moderate, and walking activities lasting at least 10 continuous minutes was 1 day/week (IQR: 1–3), 3 days/week (IQR: 2–5), and 6 days/week (IQR: 3–7), respectively. Corresponding values in the older group were 0 days/week (IQR: 0–2), 2 days/week (IQR: 1–5), and 4 days/week (IQR: 2–7). Median sedentary time in the young group was 8 h/day (IQR: 6 h 11 min–10 h) on weekdays and 6 h/day (IQR: 4 h 11 min–7 h) on weekend days. In the older group, median sedentary time was 6 h/day (IQR: 5–8 h) on both weekdays and weekend days.

Table 2 presents the descriptive characteristics of the participants with a velocity deficit who were included in the intervention following baseline assessment ($n = 57$) and randomized to the IND ($n = 28$) and NON-IND ($n = 29$) training groups.

3.2 | Training Compliance, Experience and Preference

Figure 2 summarizes participant flow throughout the study. Of the 57 participants who began the intervention, three withdrew between the PRE and MID assessments. In the IND group, one participant discontinued due to worsening gluteal discomfort that had been present before study enrollment but had initially not interfered with training. In the NON-IND group, two participants withdrew: one due to non-study-related health problems and one due to upper leg pain. The latter participant had reported no pre-existing musculoskeletal conditions at baseline. These three participants were excluded from all analyses, resulting in a sample of 54 participants for the PRE-to-MID analyses.

Of the 54 remaining participants, two were excluded from the MID-to-POST analyses. One participant, who exhibited a velocity deficit throughout the intervention and had been randomized to the NON-IND (high-load) group, inadvertently received IND (low-load) training between MID and POST because of an error in training load allocation. A second participant in the NON-IND group withdrew due to knee pain and had reported no pre-existing musculoskeletal conditions at baseline. For both participants, PRE-to-MID data were retained, whereas MID-to-POST data were excluded. Consequently, the final sample for the MID-to-POST analyses comprised 52 participants.

Adherence was calculated separately for each study period and was defined as the proportion of completed training sessions relative to those prescribed among participants included in the respective analyses. Adherence was high in both groups, exceeding 90% for all participants. When asked to rate their

TABLE 1 | Descriptive characteristics and force-velocity profile parameters by age group and sex.

	Female			Male		
	Young ($n = 30$)	**Old ($n = 30$)**	***p***	**Young ($n = 30$)**	**Old ($n = 50$)**	***p***
Age (years)	23.9 ± 2.4	70.6 ± 4.3	**<0.001**	24.6 ± 3.2	72.0 ± 3.8	**<0.001**
Body mass (kg)	58.3 ± 7.9	68.2 ± 10.4	**<0.001**	78.3 ± 9.7	78.6 ± 11.9	0.906
Height (m)	1.65 ± 0.05	1.63 ± 0.06	0.219	1.79 ± 0.06	1.76 ± 0.07	**0.023**
BMI ($kg \cdot m^{-2}$)	21.4 ± 2.5	25.6 ± 4.2	**<0.001**	24.5 ± 2.9	25.4 ± 3.1	0.169
F-V parameters						
$P_{max} \cdot BM^{-1}$ ($W \cdot kg^{-1}$)	6.2 ± 0.9	3.9 ± 1.0	**<0.001**	9.6 ± 1.6	5.6 ± 1.2	**<0.001**
$F_0 \cdot BM^{-1}$ ($N \cdot kg^{-1}$)	12.5 ± 1.9	9.6 ± 2.1	**<0.001**	16.8 ± 2.8	12.0 ± 2.3	**<0.001**
V_0 ($m \cdot s^{-1}$)	1.98 ± 0.13	1.60 ± 0.14	**<0.001**	2.28 ± 0.13	1.85 ± 0.18	**<0.001**

Note: Data are reported as mean ± SD. *p*-values indicate between-age group differences within each sex, assessed by independent *t*-tests. Bold values indicate statistical significance ($p < 0.05$).
Abbreviations: BM, body mass; BMI, body mass index; F_0, estimated maximal isometric force; F-V, force-velocity; P_{max}, estimated maximal power; V_0, estimated maximal velocity.

TABLE 2 | Descriptive characteristics for participants in the individualized (IND) and non-individualized (NON-IND) training groups.

	IND (17♂; 11♀)	NON-IND (19♂; 10♀)	*p*
Age (years)	71.8 ± 3.8	71.7 ± 4.1	0.928
Body mass (kg)	72.9 ± 13.8	74.7 ± 11.1	0.598
Height (m)	1.70 ± 0.09	1.72 ± 0.08	0.551
BMI ($kg \cdot m^{-2}$)	25.1 ± 3.9	25.4 ± 3.3	0.786
Education (*n*)			1.000
Primary education	0	1	
Secondary education	6	5	
Bachelor's or Master's degree	22	23	
Chronic conditions (*n*)			
Conditions per participant	0.9 ± 1.1	0.8 ± 0.9	0.784
Hypercholesterolemia	7	9	0.770
Hypertension	9	10	1.000
Diabetes	3	1	0.352
Thyroid problems	2	0	0.237
Gastro-intestinal problems	1	1	1.000
Cardiac arrhythmia	1	0	0.491
Pulmonary problems	0	1	1.000
Kidney disease	1	0	0.491

Note: Continuous variables are reported as mean ± SD. Continuous variables were compared between groups using independent *t*-tests or Mann–Whitney *U* tests if normality (Shapiro–Wilk) was not met. Categorical variables were compared using Fisher's exact test. Bold values indicate statistical significance ($p < 0.05$).
Abbreviation: BMI, body mass index.

training experience on a 0–10 scale, all participants who completed the intervention reported high levels of both enjoyability and feasibility (both 9.2 ± 1.0). Regarding training preference, 45% of participants (50% IND, 50% NON-IND) indicated they would have chosen low-load power training, 20% (55% IND, 45% NON-IND) preferred high-load power training, and 35% (50% IND, 50% NON-IND) reported no preference.

3.3 | Training Groups and Progression

Based on the baseline test, all participants had a velocity deficit at baseline. Therefore, in the first part of the intervention (PRE–MID), IND trained at 40% 1-RM and NON-IND at 80% 1-RM. During the mid-test, F-V profiles were re-evaluated and absolute and relative loads were adjusted for the second part of the intervention (MID–POST). Specifically, participants in IND who had a neutral profile at MID ($n = 4$) started training one session at 40% and one at 80% 1-RM. Participants in IND who had a force deficit at MID ($n = 1$) started training at 80% 1-RM. Participants in NON-IND that had a neutral profile at MID ($n = 2$) continued at 80% 1-RM. Participants in NON-IND that had a force deficit ($n = 1$) at MID switched to 40% 1-RM.

Table 3 summarizes the power training data for the IND and NON-IND training groups before (weeks 3 to 6) and after (weeks 7 to 12) the mid-test. Absolute load, relative load, and the number of repetitions differed significantly between the IND and NON-IND groups ($p < 0.001$). All velocity-related variables also differed significantly between groups during both training periods (all $p < 0.001$). In contrast, the set volume index did not differ between groups during either training period (PRE–MID: $p = 0.340$; MID–POST: $p = 0.552$), and neither did the session volume index (PRE–MID: $p = 0.326$; MID–POST: $p = 0.591$). Between the two training periods, absolute and relative load increased in the IND group ($p = 0.006$ and $p = 0.005$, respectively). Session volume index increased significantly in both groups (both $p < 0.001$).

3.4 | Leg Press Outcomes

All leg press outcomes included data from 54 participants at PRE and MID (IND: $n = 27$; NON-IND: $n = 27$) and from 52 participants at POST (IND: $n = 27$; NON-IND: $n = 25$). No outliers were identified for any leg press outcome. At baseline, the IND and NON-IND groups did not differ significantly in any leg press variable (all $p \geq 0.432$). In the time-only model, model-estimated time contrasts showed significant increases in F_0, V_0, P_{max}, and 1-RM from PRE to MID ($p < 0.001$ to $p = 0.018$). Slope was the only outcome that did not change significantly during this period ($p = 0.959$). From MID to POST, significant improvements were observed for all outcomes except V_0 ($p = 0.329$). From PRE to POST, all outcomes improved significantly ($p < 0.001$ to $p = 0.040$).

TABLE 3 | Power-training characteristics during the PRE–MID and MID–POST intervention periods in the individualized (IND) and non-individualized (NON-IND) training groups.

		IND	NON-IND	*p*
PRE-MID	Absolute load (kg)	29.4 ± 8.0	54.6 ± 14.4	**< 0.001**
	Relative load (%1-RM)	40.1 ± 0.9	79.4 ± 2.0	**< 0.001**
	Repetitions (*n*)	14.5 ± 5.0	6.8 ± 1.2	**< 0.001**
	V_{max} (m·s^{-1})	1.15 ± 0.07	0.68 ± 0.10	**< 0.001**
	V_{min} (m·s^{-1})	0.99 ± 0.06	0.56 ± 0.09	**< 0.001**
	Velocity loss (%)	14.55 ± 1.63	17.39 ± 2.68	**< 0.001**
	Set volume index (a.u.)	580.2 ± 194.9	539.8 ± 95.9	0.340
	Session volume index (a.u.)	1162.7 ± 389.8	1079.6 ± 191.8	0.326
MID-POST	Absolute load (kg)	34.9 ± 14.8*	57.0 ± 16.9	**< 0.001**
	Relative load (%1-RM)	46.4 ± 10.7*	78.2 ± 8.3	**< 0.001**
	Repetitions (*n*)	13.4 ± 3.2	7.5 ± 2.4	**< 0.001**
	V_{max} (m·s^{-1})	1.08 ± 0.16	0.68 ± 0.21	**< 0.001**
	V_{min} (m·s^{-1})	0.94 ± 0.15	0.57 ± 0.18	**< 0.001**
	Velocity loss (%)	13.74 ± 2.44	17.17 ± 2.02	**< 0.001**
	Set volume index (a.u.)	590.9 ± 105.5	571.7 ± 123.7	0.552
	Session volume index (a.u.)	1751.6 ± 321.9*	1698.8 ± 377.3*	0.591

Note: Data are reported as mean ± SD. *p*-values indicate between-group differences assessed by independent *t*-tests, with bold values indicating statistical significance ($p < 0.05$). Asterisks (*) indicate significant within-group changes between the PRE-MID and MID-POST training periods assessed by paired *t*-tests ($p < 0.01$). During PRE-MID, the IND group trained at 40% 1-RM and the NON-IND group at 80% 1-RM. During MID-POST, training loads were based on force-velocity profile characteristics assessed at MID (see Methods). V_{max} was the fastest repetition within a set. V_{min} was the slower of the two consecutive repetitions exceeding the velocity-loss threshold. Velocity loss = $([V_{max} - V_{min}]/V_{max}) \times 100$. Set volume index = repetitions × relative load (%1-RM) per set, averaged across completed sets. Session volume index = sum of set volumes per session, averaged across completed sessions.

Table 4 and Figure 4 present the within-group changes in leg press outcomes for the IND and NON-IND groups and the corresponding Time × Group contrasts, calculated as the within-group change in the IND group minus the corresponding within-group change in the NON-IND group.

Both groups demonstrated significant increases in absolute P_{max} and relative P_{max} (P_{max}·BM^{-1}) from PRE to MID (all $p \leq 0.001$, ES = 0.19–0.23). No significant changes were observed from MID to POST (all $p \geq 0.281$, ES = 0.07–0.08), resulting in significant increases from PRE to POST (all $p < 0.001$, ES = 0.26–0.30). No significant between-group differences in change were observed for any time contrast (all $p \geq 0.557$).

For force-related outcomes, the NON-IND group demonstrated significant increases in absolute F_0 and relative F_0 (F_0·BM^{-1}) from PRE to MID ($p = 0.017$, ES = 0.21; $p = 0.004$, ES = 0.24, respectively) and from PRE to POST (both $p < 0.001$; ES = 0.33 and 0.39, respectively). In contrast, no significant changes were observed in the IND group across any time contrast (all $p \geq 0.100$). Accordingly, significant between-group differences in change were observed from PRE to POST in favor of the NON-IND group for both F_0 ($\Delta = -48.6$ N, 95% CI −95.1 to −2.2, $p = 0.040$) and F_0·BM^{-1} ($\Delta = -0.7$ N·kg^{-1}, 95% CI −1.3 to −0.1, $p = 0.022$).

For V_0, the IND group improved from PRE to MID and PRE to POST (both $p \leq 0.001$; ES = 0.52–0.54), whereas no significant changes were observed in the NON-IND group across any time contrast (all $p \geq 0.526$; ES = −0.15 to 0.16). Consequently, a significant between-group difference in change was observed from PRE to POST in favor of the IND group ($\Delta = 0.08$ m·s^{-1}, 95% CI 0.02 to 0.14, $p = 0.009$).

For slope, the NON-IND group demonstrated a significant shift towards a more negative value from PRE to POST ($\Delta = -48.5$ N·s·m^{-1}, $p = 0.004$, ES = −0.36), indicating a greater gain in F_0 compared to V_0. In contrast, no significant changes were observed in the IND group across any time contrast (all $p \geq 0.591$; ES = −0.13 to 0.18). Consequently, a significant between-group difference in change was observed from PRE to POST in favor of the NON-IND group ($\Delta = 51.8$ N·s·m^{-1}, 95% CI 11.0 to 92.7, $p = 0.013$).

For 1-RM, only the NON-IND group improved significantly from PRE to MID ($p = 0.001$, ES = 0.22). However, significant increases from MID to POST were observed in both groups (IND: $p = 0.038$, ES = 0.12; NON-IND: $p = 0.007$, ES = 0.16), resulting in significant PRE-to-POST improvements in both groups (IND: $p = 0.001$, ES = 0.17; NON-IND: $p < 0.001$, ES = 0.36). Significant between-group differences in change favored the NON-IND group from PRE to MID ($\Delta = -3.0$ kg, 95% CI −6.0 to −0.0, $p = 0.049$) and from PRE to POST ($\Delta = -3.7$ kg, 95% CI −6.8 to −0.7, $p = 0.017$).

TABLE 4 | Model-estimated marginal means for leg press outcomes at pre-, mid- and post-intervention in the individualized and non-individualized groups, with within-group changes and Time × Group contrasts.

Variable	Test	IND	NON-IND	Contrast	IND change	*p*	ES	NON-IND change	*p*	ES	Time × Group contrast	*p*
P_{max} (W)	PRE	371.8 (325.4, 418.3)	366.7 (320.2, 413.2)	PRE–MID	22.9 (8.5, 37.2)	**0.001**	0.19	26.1 (11.7, 40.5)	**< 0.001**	0.22	−3.2 (−19.8, 13.4)	0.701
	MID	394.7 (348.2, 441.2)	392.8 (346.3, 439.2)	MID–POST	9.1 (−5.3, 23.5)	0.375	0.07	9.7 (−5.1, 24.5)	0.345	0.07	−0.5 (−17.4, 16.3)	0.951
	POST	403.8 (357.4, 450.3)	402.4 (355.8, 449.0)	PRE–POST	32.0 (17.6, 46.4)	**< 0.001**	0.26	35.7 (20.9, 50.6)	**< 0.001**	0.27	−3.7 (−20.6, 13.1)	0.660
P_{max}·BM^{-1} (W·kg^{-1})	PRE	5.12 (4.55, 5.69)	4.95 (4.38, 5.52)	PRE–MID	0.32 (0.15, 0.50)	**< 0.001**	0.22	0.36 (0.18, 0.53)	**< 0.001**	0.23	−0.04 (−0.24, 0.17)	0.731
	MID	5.45 (4.87, 6.02)	5.31 (4.77, 5.88)	MID–POST	0.10 (−0.08, 0.27)	0.509	0.07	0.13 (−0.05, 0.31)	0.281	0.08	−0.03 (−0.23, 0.18)	0.803
	POST	5.55 (4.97, 6.12)	5.43 (4.89, 6.01)	PRE–POST	0.42 (0.25, 0.60)	**< 0.001**	0.30	0.48 (0.30, 0.66)	**< 0.001**	0.30	−0.06 (−0.27, 0.14)	0.557
F_0 (N)	PRE	863.8 (776.4, 951.3)	845.4 (757.9, 932.8)	PRE–MID	10.1 (−29.7, 49.8)	1.000	0.05	46.2 (6.4, 85.9)	**0.017**	0.21	−36.1 (−81.9, 9.7)	0.121
	MID	873.9 (786.4, 961.3)	891.5 (804.1, 979.0)	MID–POST	23.8 (−15.9, 63.6)	0.442	0.10	36.4 (−4.6, 77.3)	0.098	0.15	−12.5 (−59.0, 34.0)	0.594
	POST	897.7 (810.3, 985.2)	927.9 (840.1, 1015.7)	PRE–POST	33.9 (−5.9, 73.6)	0.121	0.15	82.5 (41.6, 123.4)	**< 0.001**	0.33	−48.6 (−95.1, −2.2)	**0.040**
F_0·BM^{-1} (N·kg^{-1})	PRE	11.9 (10.9, 13.0)	11.4 (10.3, 12.5)	PRE–MID	0.2 (−0.3, 0.7)	1.000	0.07	0.7 (0.2, 1.2)	**0.004**	0.24	−0.5 (−1.1, 0.1)	0.098
	MID	12.1 (11.1, 13.2)	12.1 (11.0, 13.2)	MID–POST	0.3 (−0.3, 0.8)	0.639	0.09	0.5 (−0.1, 1.0)	0.096	0.15	−0.2 (−0.8, 0.4)	0.495
	POST	12.4 (11.3, 13.5)	12.6 (11.5, 13.7)	PRE–POST	0.5 (−0.1, 1.0)	0.100	0.17	1.2 (0.6, 1.7)	**< 0.001**	0.39	−0.7 (−1.3, −0.1)	**0.022**
V_0 (m·s^{-1})	PRE	1.70 (1.63, 1.76)	1.71 (1.64, 1.78)	PRE–MID	0.09 (0.03, 0.14)	**< 0.001**	0.54	0.03 (−0.02, 0.08)	0.526	0.16	0.06 (0.00, 0.12)	0.063
	MID	1.78 (1.72, 1.85)	1.74 (1.67, 1.81)	MID–POST	0.00 (−0.06, 0.05)	1.000	−0.03	−0.03 (−0.08, 0.03)	0.576	−0.15	0.03 (−0.04, 0.09)	0.420
	POST	1.78 (1.71, 1.85)	1.71 (1.64, 1.78)	PRE–POST	0.08 (0.03, 0.14)	**0.001**	0.52	0.00 (−0.05, 0.05)	1.000	0.01	0.08 (0.02, 0.14)	**0.009**
Slope (N·s·m^{-1})	PRE	−505.6 (−550.4, −460.8)	−492.0 (−536.8, −447.2)	PRE–MID	18.6 (−16.3, 53.6)	0.591	0.18	−19.7 (−54.7, 15.2)	0.519	−0.17	38.3 (−1.9, 78.6)	0.062
	MID	−487.0 (−531.8, −442.2)	−511.7 (−556.5, −466.9)	MID–POST	−15.3 (−50.2, 19.7)	0.869	−0.13	−28.8 (−64.7, 7.2)	0.163	−0.22	13.5 (−27.4, 54.3)	0.515
	POST	−502.3 (−547.1, −457.5)	−540.5 (−585.8, −495.2)	PRE–POST	3.4 (−31.6, 38.3)	1.000	0.03	−48.5 (−84.4, −12.5)	**0.004**	−0.36	51.8 (11.0, 92.7)	**0.013**
1-RM (kg)	PRE	73.1 (65.2, 81.0)	68.7 (60.8, 76.6)	PRE–MID	1.2 (−1.4, 3.7)	0.851	0.05	4.1 (1.6, 6.7)	**0.001**	0.22	−3.0 (−6.0, −0.0)	**0.049**
	MID	74.3 (66.4, 82.2)	72.9 (65.0, 80.7)	MID–POST	2.7 (0.1, 5.3)	**0.038**	0.12	3.4 (0.8, 6.1)	**0.007**	0.16	−0.7 (−3.8, 2.3)	0.639
	POST	77.0 (69.1, 84.9)	76.3 (68.4, 84.2)	PRE–POST	3.9 (1.3, 6.4)	**0.001**	0.17	7.6 (4.9, 10.2)	**< 0.001**	0.36	−3.7 (−6.8, −0.7)	**0.017**

Note: Values are estimated marginal means (EMMs) presented with 95% confidence intervals (CIs). Time × Group contrasts (i.e., between-group differences in change) were calculated as the within-group change in the IND group minus the within-group change in the NON-IND group. Bold values indicate statistical significance ($p < 0.05$).

Abbreviations: 1-RM, one-repetition maximum; BM, body mass; ES, effect size (Hedges' g_{av}); F_0, estimated maximal isometric force; IND, individualized; MID, mid-intervention; NON-IND, non-individualized; P_{max}, estimated maximal power; POST, post-intervention; PRE, pre-intervention; V_0, estimated maximal velocity.

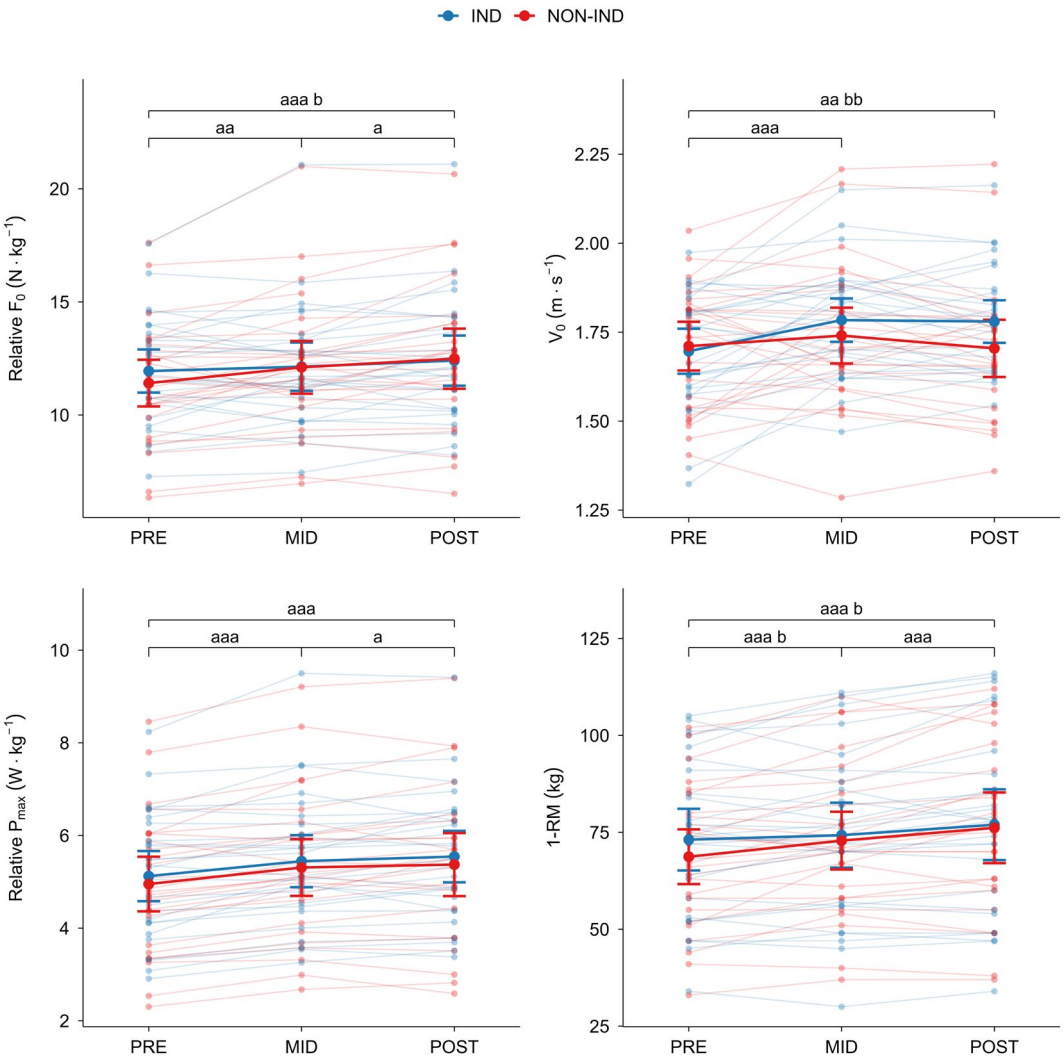


FIGURE 4 | Changes in leg press outcomes from pre- to mid- and post-intervention. Estimated relative maximal muscle force (F_0 normalized to body mass), estimated maximal movement velocity (V_0), estimated relative maximal muscle power (P_{max} normalized to body mass) and one-repetition maximum (1-RM) are shown for the individualized (IND; blue) and non-individualized (NON-IND; red) groups. Individual trajectories are displayed alongside group estimated marginal means ±95% confidence intervals. Lines connecting group means illustrate changes over time. Brackets indicate pairwise comparisons between assessment time points. The letter *a* denotes a significant time effect, whereas *b* denotes a significant Time × Group interaction. One, two or three letters indicate significance at $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively.

3.5 | Functional Performance

The functional performance analyses were based on slightly different sample sizes across outcomes due to missing assessments. For most outcomes, data were available for 54 participants at PRE and MID (IND: $n = 27$; NON-IND: $n = 27$) and 52 participants at POST (IND: $n = 27$; NON-IND: $n = 25$). Stair-climb outcomes were available for fewer participants because of processing errors, with sample sizes of $n = 54$ (IND: $n = 27$; NON-IND: $n = 27$) at PRE, $n = 53$ (IND: $n = 26$; NON-IND: $n = 27$) at MID, and $n = 48$ (IND: $n = 24$; NON-IND: $n = 24$) at POST. For the CMJ, one participant in the IND group did not perform the test at any assessment to avoid aggravating a pre-existing Achilles tendon problem, resulting in a sample size of $n = 26$ for the IND group across all time points. Two outliers were identified among the functional performance outcomes: one for 5×STS power at MID and one for 5×STS duration at PRE. Excluding these observations did not affect statistical significance or alter the study conclusions.

At baseline, no significant differences were observed between the IND and NON-IND groups for any functional performance outcome (all $p \geq 0.229$). In the time-only model, model-estimated contrasts indicated significant improvements from PRE to MID for all functional outcomes ($p < 0.001$), except stair ascent power ($p = 0.496$). No significant changes were observed from MID to POST (all $p \geq 0.138$), resulting in significant improvements from PRE to POST for all functional outcomes ($p \leq 0.003$), except stair ascent power ($p = 0.172$).

Table 5 presents group-specific changes in functional performance and the corresponding between-group differences in change over time (Time × Group interactions). Between-group contrasts in change revealed no significant differences between

TABLE 5 | Model-estimated marginal means for functional outcomes at pre-, mid- and post-intervention in the individualized and non-individualized groups, with within-group changes and Time × Group contrasts.

Variable	Test	IND	NON-IND	Contrast	IND change	*p*	ES	NON-IND change	*p*	ES	Time × Group contrast	*p*
HGS (kg)	PRE	33.8 (30.4, 37.2)	35.2 (31.8, 38.5)	PRE–MID	2.1 (0.5, 3.6)	**0.004**	0.25	1.6 (0.1, 3.2)	**0.033**	0.17	0.4 (−1.3, 2.2)	0.619
	MID	35.9 (32.5, 39.3)	36.8 (33.4, 40.2)	MID–POST	−0.3 (−1.8, 1.3)	1.000	−0.03	−0.4 (−2.0, 1.2)	1.000	−0.03	0.1 (−1.7, 1.9)	0.886
	POST	35.6 (32.2, 39.0)	36.4 (33.0, 39.8)	PRE–POST	1.8 (0.3, 3.4)	**0.014**	0.21	1.2 (−0.3, 2.8)	0.175	0.11	0.6 (−1.2, 2.4)	0.527
10 m MGS (s)	PRE	4.50 (4.23, 4.78)	4.74 (4.47, 5.02)	PRE–MID	−0.31 (−0.48, −0.14)	**< 0.001**	−0.44	−0.36 (−0.53, −0.20)	**< 0.001**	−0.44	0.05 (−0.14, 0.25)	0.579
	MID	4.20 (3.92, 4.47)	4.38 (4.10, 4.65)	MID–POST	0.00 (−0.16, 0.17)	1.000	0.01	0.01 (−0.16, 0.19)	1.000	0.02	−0.01 (−0.21, 0.18)	0.916
	POST	4.20 (3.92, 4.47)	4.39 (4.12, 4.67)	PRE–POST	−0.31 (−0.47, −0.14)	**< 0.001**	−0.48	−0.35 (−0.52, −0.18)	**< 0.001**	−0.42	0.04 (−0.15, 0.24)	0.659
TUG (s)	PRE	5.92 (5.59, 6.25)	6.10 (5.77, 6.44)	PRE–MID	−0.22 (−0.43, −0.01)	**0.036**	−0.26	−0.36 (−0.57, −0.15)	**< 0.001**	−0.36	0.14 (−0.1, 0.38)	0.248
	MID	5.70 (5.37, 6.03)	5.74 (5.41, 6.08)	MID–POST	0.07 (−0.14, 0.28)	1.000	0.10	0.00 (−0.22, 0.21)	1.000	−0.02	0.07 (−0.17, 0.32)	0.559
	POST	5.77 (5.43, 6.1)	5.74 (5.40, 6.08)	PRE–POST	−0.15 (−0.36, 0.06)	0.246	−0.18	−0.36 (−0.58, −0.15)	**< 0.001**	−0.34	0.21 (−0.03, 0.46)	0.086
CMJ (mm)	PRE	159.8 (137.9, 181.8)	155.6 (134.0, 177.1)	PRE–MID	10.7 (3.7, 17.7)	**0.001**	0.18	14.2 (7.3, 21.0)	**< 0.001**	0.25	−3.5 (−11.5, 4.6)	0.395
	MID	170.5 (148.6, 192.4)	169.7 (148.2, 191.2)	MID–POST	1.7 (−5.3, 8.8)	1.000	0.03	4.5 (−2.6, 11.6)	0.369	0.08	−2.8 (−10.9, 5.3)	0.496
	POST	172.2 (150.3, 194.2)	174.2 (152.7, 195.8)	PRE–POST	12.4 (5.4, 19.5)	**< 0.001**	0.23	18.7 (11.6, 25.8)	**< 0.001**	0.32	−6.3 (−14.4, 1.9)	0.130
SA duration (s)	PRE	1.83 (1.7, 1.96)	1.81 (1.69, 1.94)	PRE–MID	−0.08 (−0.14, −0.01)	**0.011**	−0.21	−0.07 (−0.13, −0.01)	**0.028**	−0.19	−0.01 (−0.08, 0.06)	0.794
	MID	1.76 (1.63, 1.88)	1.75 (1.62, 1.88)	MID–POST	0.00 (−0.07, 0.06)	1.000	0.02	0.03 (−0.03, 0.1)	0.651	0.10	−0.04 (−0.11, 0.04)	0.349
	POST	1.75 (1.62, 1.88)	1.78 (1.65, 1.91)	PRE–POST	−0.08 (−0.14, −0.01)	**0.011**	−0.25	−0.03 (−0.1, 0.03)	0.600	−0.10	−0.05 (−0.12, 0.03)	0.235
SA power ($W \cdot kg^{-1}$)	PRE	8.07 (7.26, 8.88)	7.92 (7.12, 8.73)	PRE–MID	−0.01 (−0.51, 0.49)	1.000	−0.01	0.20 (−0.29, 0.69)	0.959	0.09	−0.21 (−0.78, 0.36)	0.465
	MID	8.06 (7.25, 8.87)	8.13 (7.32, 8.93)	MID–POST	0.29 (−0.22, 0.8)	0.516	0.14	−0.08 (−0.59, 0.44)	1.000	−0.04	0.37 (−0.22, 0.95)	0.222
	POST	8.35 (7.54, 9.17)	8.05 (7.24, 8.86)	PRE–POST	0.28 (−0.23, 0.79)	0.557	0.13	0.13 (−0.39, 0.64)	1.000	0.08	0.16 (−0.43, 0.74)	0.603

(Continues)

TABLE 5 | (Continued)

Variable	Test	IND	NON-IND	Contrast	IND change	*p*	ES	NON-IND change	*p*	ES	Time × Group contrast	*p*
5×STS duration (s)	PRE	9.19 (8.48, 9.89)	9.29 (8.58, 9.99)	PRE–MID	−1.14 (−1.61, −0.67)	**< 0.001**	−0.69	−0.99 (−1.46, −0.51)	**< 0.001**	−0.44	−0.15 (−0.70, 0.39)	0.576
	MID	8.05 (7.34, 8.75)	8.30 (7.60, 9.01)	MID–POST	−0.17 (−0.64, 0.31)	1.000	−0.12	0.11 (−0.38, 0.59)	1.000	0.05	−0.27 (−0.83, 0.28)	0.329
	POST	7.88 (7.17, 8.58)	8.41 (7.70, 9.12)	PRE–POST	−1.31 (−1.78, −0.83)	**< 0.001**	−0.81	−0.88 (−1.37, −0.39)	**< 0.001**	−0.37	−0.43 (−0.98, 0.13)	0.128
5×STS power ($W \cdot kg^{-1}$)	PRE	3.66 (3.43, 3.90)	3.72 (3.48, 3.96)	PRE–MID	0.36 (0.16, 0.56)	**< 0.001**	0.65	0.32 (0.12, 0.52)	**< 0.001**	0.46	0.04 (−0.19, 0.27)	0.737
	MID	4.02 (3.79, 4.26)	4.04 (3.80, 4.28)	MID–POST	0.12 (−0.08, 0.32)	0.407	0.22	0.05 (−0.16, 0.25)	1.000	0.08	0.08 (−0.15, 0.31)	0.508
	POST	4.15 (3.91, 4.38)	4.08 (3.84, 4.32)	PRE–POST	0.48 (0.28, 0.68)	**< 0.001**	0.83	0.37 (0.16, 0.57)	**< 0.001**	0.51	0.12 (−0.12, 0.35)	0.322
30sSTS repetitions (*n*)	PRE	15.3 (14.3, 16.4)	15.2 (14.2, 16.3)	PRE–MID	1.7 (1.0, 2.4)	**< 0.001**	0.61	1.2 (0.5, 1.8)	**< 0.001**	0.42	0.5 (−0.3, 1.3)	0.194
	MID	17.0 (16.0, 18.0)	16.4 (15.3, 17.4)	MID–POST	0.3 (−0.4, 0.9)	1.000	0.09	0.0 (−0.7, 0.7)	1.000	0.03	0.2 (−0.6, 1.0)	0.564
	POST	17.3 (16.2, 18.3)	16.4 (15.3, 17.5)	PRE–POST	1.9 (1.2, 2.6)	**< 0.001**	0.66	1.2 (0.5, 1.9)	**< 0.001**	0.39	0.8 (−0.1, 1.6)	0.065
30sSTS power ($W \cdot kg^{-1}$)	PRE	3.44 (3.2, 3.68)	3.26 (3.02, 3.50)	PRE–MID	0.22 (0.03, 0.41)	**0.014**	0.35	0.26 (0.07, 0.44)	**0.003**	0.39	−0.04 (−0.25, 0.18)	0.736
	MID	3.66 (3.42, 3.90)	3.52 (3.28, 3.76)	MID–POST	−0.01 (−0.19, 0.18)	1.000	−0.02	0.11 (−0.08, 0.30)	0.452	0.17	−0.12 (−0.34, 0.09)	0.265
	POST	3.65 (3.41, 3.89)	3.63 (3.39, 3.88)	PRE–POST	0.21 (0.03, 0.40)	**0.020**	0.37	0.37 (0.18, 0.56)	**< 0.001**	0.52	−0.16 (−0.38, 0.06)	0.149

Note: Values are estimated marginal means (EMMs) presented with 95% confidence intervals (CIs). Time×Group contrasts (i.e., between-group differences in change) were calculated as the within-group change in the IND group minus the within-group change in the NON-IND group. Bold values indicate statistical significance ($p < 0.05$).

Abbreviations: 30sSTS, 30 s sit-to-stand; 5×STS, 5-repetition sit-to-stand; CMJ, countermovement jump; ES, effect size (Hedges' g_{av}); HGS, handgrip strength; IND, individualized; MGS, maximal gait speed; MID, mid-intervention; NON-IND, non-individualized; POST, post-intervention; PRE, pre-intervention; SA, stair ascent (6 steps); TUG, timed up-and-go.

the IND and NON-IND groups for any functional performance outcome at any time interval (all $p \geq 0.065$).

4 | Discussion

In older adults classified as velocity-deficient relative to a young reference group, training with loads matched to the predominant F-V deficit did not result in greater improvements in maximal muscle power or functional performance than training with loads deliberately not matched to that deficit. The two approaches did, however, lead to different F-V adaptations: the IND group (which primarily trained at low loads) improved in their maximal velocity capacity, whereas the NON-IND group (which primarily trained at high loads) improved their maximal force capacity.

4.1 | Defining Deficits

Velocity-deficit classifications were predominant in this sample under the present young reference z-score method. This pattern is in line with recent longitudinal research [1] that also identified a reduced ability to generate force at high contraction velocities as a hallmark of aging. From a physiological standpoint, potential mechanisms underlying the preferential decline in force production at high velocities include the selective atrophy and loss of Type II muscle fibers [23], reductions in fiber shortening velocity [24] and impairments in neural activation during the early phase of muscle contraction [25].

However, although the present classification pattern is broadly consistent with previous longitudinal observations suggesting a greater age-related decline in V_0 than F_0 [1], this interpretation remains speculative and warrants further investigation. Approximately one-quarter of participants were classified as having balanced profiles or force deficits. Moreover, the velocity-deficit group showed substantial heterogeneity, with a wide range of deficit magnitudes. This variability suggests that many participants lay on a continuum rather than in clearly separated profile categories, so their classification depends on the reference sample, z-score scaling and cut-off applied. The proportion of participants classified as velocity-deficient should therefore be interpreted as specific to the present operational method and not as an estimate of the prevalence of velocity deficits in older adults.

Defining F-V deficits in older adults remains a key methodological challenge, as no consensus exists for classification. Unlike young athletes, whose deficits can be defined relative to mathematically derived optimal F-V profiles [10, 12], no such benchmark exists for older adults. Consequently, researchers have employed divergent approaches. Alcazar et al. [9] used tertile-based cut-offs. Lindberg et al. [13] compared each participant's profile against a reference line defined by the median F-V slope of their sample. Calaway et al. [26] anchored a reference line to the highest maximal force and velocity observed within their sample and categorized participants with steeper or less steep slopes compared to the reference line in velocity or force deficit groups, respectively. The present study took a conceptually distinct approach by benchmarking older adults against a young reference group rather than against one another.

The consequences of these methodological differences are substantial. While Alcazar et al. [9] and Lindberg et al. [13] reported deficit prevalences that are mathematically fixed by their use of tertile and median splits, respectively, Calaway et al. [26] and the present study derived prevalence empirically. Interestingly, Calaway et al. [26] classified 24% as velocity-deficit and 76% as force-deficit, while our approach classified 5% of participants as having a force deficit and 73.75% a velocity deficit. These discrepancies demonstrate that the choice of classification method strongly shapes the perceived distribution of deficit types, which, by extension, has strong implications for the design of F-V-guided training programs.

We argue that benchmarking against a young reference group provides a more meaningful standard than within-sample comparisons, as it better captures differences attributable to aging rather than sample-specific variation. However, the use of a young reference group also has limitations. Although young adults were screened to exclude individuals who regularly participated in resistance training or competitive sports, and information on the frequency of moderate and vigorous physical activity was collected, more detailed information regarding the specific types of recreational activities performed was not recorded. Therefore, we cannot exclude the possibility that some participants were more physically active than the general young adult population, which may have influenced the derived reference values and, consequently, the classification of force and velocity deficits. Nevertheless, the screening procedures likely reduced the influence of highly trained individuals on the reference values.

The use of z-scores to compare older and younger adults also warrants consideration. First, the threshold of a ≥ 1.0 difference between the velocity and force z-scores used to classify participants into deficit groups was selected based on exploratory analyses of previously collected leg press data. This threshold was selected to ensure meaningful differences between z-score-based force- and velocity-deficit groups. As it represents an operational classification rule rather than a validated physiological cut-point, the terms 'force deficit' and 'velocity deficit' should be interpreted as study-specific classifications rather than physiological diagnoses. Sensitivity analyses using alternative cut-offs (0.5, 1.0, and 1.5 z-score units) yielded a similar overall pattern of results. At the chosen cut-off of 1.0, 4 participants were classified as force-deficient, 59 as velocity-deficient, and 17 as neutral. At cut-offs of 0.5 and 1.5, the corresponding force-deficit/velocity-deficit/neutral counts were 8/62/10 and 1/41/38, respectively. Across all cut-offs, velocity-deficient profiles remained the most prevalent. Increasing the cut-off primarily shifted participants to the neutral category, resulting in fewer force- and velocity-deficient classifications.

Second, z-scores were used because they express group differences relative to sample variability, enabling scale-independent and interpretable comparisons of effect magnitude. This was considered an advantage over percentage differences, which do not account for variability and are therefore less informative for between-group comparisons. Nevertheless, z-scores depend

not only on the difference between older and younger adults but also on the variability within the young reference group. In the present study, variability relative to the mean was substantially greater for relative F_0 (16.7% in males and 15.2% in females) than for V_0 (5.7% and 6.6%, respectively). As a result, the same relative age-related difference produced a smaller (i.e., less negative) z-score for F_0 than for V_0, which mathematically increases the likelihood of a participant being classified as velocity-deficient rather than force-deficient. The observed prevalence of velocity deficits (73.8%) therefore partly reflects the statistical properties of the young reference distribution, and should not be interpreted as direct evidence that most older adults have a physiological velocity deficit.

The F-V slope could provide a useful complementary approach to comparing F_0 and V_0 deficits separately because it summarizes the F-V relationship in a single descriptor, so that classification can rely on a single standardized outcome instead of two. The slope is not intrinsically scale-independent, as its value depends on the measurement units and normalization procedure used. Standardizing the slope against a reference sample would make it unit-free, as is the case for the z-scores of F_0 and V_0 used in the present study. However, this would not remove the dependencies shared by both approaches, namely the normalization applied beforehand (e.g., absolute versus body-mass-normalized force), the composition of the reference sample and the between-subject variability and measurement error of each parameter. In addition, because the slope is derived from both F_0 and V_0, its reliability depends on the measurement uncertainty of both parameters. Errors in either intercept propagate into the slope, and relatively small variations in F_0 or V_0 can lead to disproportionately large changes in its value, reducing its stability. In contrast, when F_0 and V_0 deficits are compared separately, measurement error remains specific to each parameter. Consistent with this reasoning, previous research has reported lower reliability for the F-V slope than for the individual intercepts [16]. We therefore did not adopt the slope in the present study. Overall, neither approach has yet been established as superior for identifying physiologically meaningful force or velocity deficits or for guiding resistance training prescription in older adults, and further research is needed.

Along with the methodological decisions regarding participant classification, the resistance modality used to derive F-V variables may represent an additional source of variability that should be considered when defining deficits. Alcazar et al. employed inertial resistance [9], whereas Lindberg et al. [13], Calaway et al. [26] and the present study all used pneumatic resistance. Previous research suggests that the mechanical characteristics of different resistance modalities can influence exercise performance and training adaptations [27]. Therefore, modality-specific differences may also affect the assessment of force and velocity capacities and, consequently, the classification of force- and velocity-deficit profiles.

4.2 | Muscle Power

Despite our different method to categorize deficits, our results align with those of Lindberg et al. [13], as neither study supports the previously suggested benefit of training based on the F-V profile [7, 9]. Instead, our findings are consistent with previous evidence showing that muscle power in older adults can improve following training at different loading intensities [6, 7], without establishing a single optimal training load [3]. Furthermore, in agreement with previous research [7], the standard deviation of maximal power gains in our study exceeded the mean improvement in both groups, indicating that training according to the F-V deficit did not attenuate the substantial interindividual variability in adaptive responses to power training among older adults.

The muscle power gains observed in our study (32.0 W; 8.6% in the IND group and 35.7 W; 9.7% in the NON-IND group) exceeded those reported by Lindberg et al. [13], where the generic and individualized training groups improved by 5% and 3.5%, respectively. This discrepancy may be related to their use of a pneumatic leg press for testing but not for training—an approach that, consistent with the principle of training specificity, often results in smaller improvements when the test does not match the training modality [28]. Compared to Rodriguez-Lopez et al. [7], power improvements in the present study were smaller, as they observed increases of 28% and 40% in their low- and high-load groups, respectively. This discrepancy may be partly attributable to differences in the training device. Rodriguez-Lopez et al. [7] used a horizontal leg press on which the effective load comprised both the added resistance and the participant's own body mass. In contrast, the pneumatic leg press employed in the present study provided a relatively constant external resistance with reduced inertial effects compared with conventional weight-stack devices. Consequently, differences in loading characteristics, limb acceleration and changes in joint moments throughout the movement may have influenced muscular and joint loading, thereby contributing to differences in training adaptations and performance outcomes between studies.

It is also possible that device characteristics played only a minor role and that variables such as load progression influenced the outcomes. When comparing the first and second halves of the present intervention, most improvements occurred during the initial 6 weeks. One explanation could be that the load progression was insufficient to drive continued improvements. However, because we re-evaluated the F-V profile and 1-RM at mid-intervention to adjust training loads based on early adaptations, a lack of progression alone may not fully account for this pattern. Instead, given that neural adaptations contribute substantially to strength gains in the early phases of training, while morphological adaptations play a larger role in later stages [29], we speculate that our observed pattern is more likely explained by this shift in adaptation mechanisms rather than by limitations in load progression.

4.3 | The Force-Velocity Profile

Although the similar power gains in the IND and NON-IND groups did not align with our initial hypothesis, they were not entirely unexpected. Most older adults in this study predominantly exhibited a velocity deficit, yet the majority also demonstrated reduced force capacities relative to the young reference group. Given this dual potential for improvement

and the distinct F-V adaptations observed in the two intervention groups, it is perhaps unsurprising that both low-load and high-load training enhanced maximal power to a similar extent. Specifically, the IND group increased V_0 without significant changes in F_0 or slope, whereas the NON-IND group increased F_0 and demonstrated a significant shift in slope towards a more negative value. Despite these differences in F-V adaptations, both groups improved P_{max}. These findings suggest that improvements in maximal power can be accompanied by different changes in the F-V profile.

However, these results also raise questions about the training phase, or the duration, over which F-V-guided training should be evaluated. The similar improvements in P_{max} observed in both groups suggest that different training stimuli may be equally effective during the initial 12 weeks of training. It remains unclear, however, whether training according to the F-V deficit would prove more effective beyond these early stages, particularly since less trained individuals typically show large improvements at the start of any new training program [30]. Exercise programming may therefore play a relatively minor role in shaping immediate adaptations to unfamiliar exercise [31], while becoming increasingly important as training progresses. Future research could also clarify whether training emphasis should shift between force-oriented and velocity-oriented modalities once progress in one component begins to plateau.

Notably, in contrast to the present study, Lindberg et al. [13] did not observe velocity-specific adaptations following low-load training. Although they mentioned that this may be attributed to age-related limitations in high-velocity adaptability [13], evidence indicated that older adults can still improve maximal velocity [7, 32], Type II fiber size and power [33], neural drive and rapid force production [7]. We therefore speculate that the absence of velocity-specific adaptations in their study may reflect the type of resistance used for training. Unlike the present study, Lindberg et al. [13] did not use pneumatic resistance. Instead, their participants were required to overcome the inertia of the external load at movement initiation [27], which may have constrained achievable movement velocity and confined training predominantly to the higher-load region of the F-V relationship.

Adding to the discussion of velocity-specific adaptations, previous studies have reported increases in maximal velocity following both low- and high-load training, while this effect was not observed in the present study. For example, Rodriguez-Lopez et al. [7] reported significant improvements in maximal velocity following high-load training. One potential explanation for the discrepant findings relates to the magnitude of velocity loss accumulated during training. In Rodriguez-Lopez et al. [7], the high-load group trained with an average velocity loss of 5.6%, whereas participants in our NON-IND (i.e., high-load) group achieved an average velocity loss of 17.3%. Although a velocity-loss threshold of 10% was prescribed in the present study, sets were terminated only after two consecutive repetitions exceeded this threshold. Velocity loss was calculated as the relative reduction from the fastest repetition in the set to the slower of these two threshold-exceeding repetitions, resulting in actual velocity losses that exceeded the nominal threshold. Previous work suggests that lower velocity losses may preferentially promote adaptations towards the velocity end of the F-V spectrum [26], whereas greater velocity losses may shift the training stimulus away from velocity-specific adaptations. Therefore, the larger velocity losses in the high-load group accumulated in the present study may have reduced the stimulus for improving maximal velocity, potentially contributing to the absence of changes in V_0.

The force-specific adaptations observed following NON-IND training align well with studies showing greater improvements in maximal force following high- versus low-load training [7, 34], with research demonstrating that heavy resistance training effectively enhances muscle strength over short-to-medium training periods [35]. This interpretation is further supported by the significant shift in slope towards a more negative value observed only in the NON-IND group, indicating a shift of the F-V profile towards greater force-producing capacity. Interestingly, earlier studies have also documented increases in maximal force following low-load training [7, 13, 36], whereas in our study the IND group did not show a significant increase in F_0. As discussed earlier, the increases in F_0 reported by Lindberg et al. [13] and Murphy et al. [36] may reflect that their low-load training occurred in a higher-force region of the F-V relationship. This rationale may also help explain discrepancies with Rodriguez-Lopez et al. [7], who used an inertial rather than a pneumatic leg press, thereby altering the mechanical demands and loading stimulus.

Notably, although no significant increase in F_0 was observed, the IND group did demonstrate a significant improvement in 1-RM. However, these outcomes should be interpreted as related but distinct indicators of neuromuscular performance, as 1-RM reflects performance during a specific dynamic strength task whereas F_0 represents the theoretical maximal force derived from the F-V relationship. The discrepancy may reflect task-specific adaptations, measurement variability, or limitations associated with estimating F_0 through extrapolation.

4.4 | Functional Performance

Although maximal muscle power appears to improve regardless of whether gains are driven by the force or velocity component, some researchers argue that velocity is particularly critical for enhancing functional ability in older adults. Greater movement velocity may be especially important for tasks demanding rapid lower-limb responses, such as crossing a street safely or recovering balance to avoid a fall [37]. It has also been identified as an independent predictor of walking performance [38], as well as chair-rise and stair-climbing ability [39], underscoring its relevance to functional independence. Notably, high-velocity resistance training appears to enhance power primarily through neural adaptations, including greater agonist muscle activation and reduced antagonist coactivation [40]. This improved coordination among muscle groups may be an important mechanism underlying the transfer of resistance training adaptations to functional performance in older adults [41].

Based on these arguments, one might expect the IND group, who demonstrated clear improvements in maximal velocity while the NON-IND group did not, to show greater gains in functional performance. However, our results do not support this assumption as there was no significant difference in the

change in functional outcomes between groups. This finding is consistent with Lindberg et al. [13], who also reported significant improvements in stair climbing, sit-to-stand performance, and grip strength, with no differences between individualized and generic power training. Together, these results suggest that training at loads matched to the predominant F-V deficit did not lead to greater improvements in functional performance over training at loads deliberately not matched to the deficit.

Several factors may explain why. First, functional tasks draw on power at different points along the F-V continuum [42]. Although gait speed appears more closely linked to lower-load power and chair-rise performance to higher-load power [39, 43], the load at which power must be produced likely varies not only by task type but also across movement phases within the same task. This complexity may limit the functional impact of shifting an individual's F-V profile in one targeted direction. Second, both groups shared an emphasis on power-oriented execution, which likely produced overlapping neuromuscular adaptations and diminished functional differences between loading strategies. Third, our sample consisted of healthy, well-functioning older adults, which may have introduced a ceiling effect whereby limited room for improvement attenuated detectable between-group differences.

Whether the observed within-group improvements are truly attributable to leg press power training also warrants careful consideration. The absence of a non-exercising control group, justified by the well-established efficacy of power training in older adults [3] and our comparative rather than efficacy-focused design, means that contributions from repeated testing or other unassessed factors cannot be excluded. Moreover, leg extensor power may only explain 12%–45% of the variance in functional performance [44]. Other factors also contribute substantially, as higher levels of physical activity and lower sedentary behavior are also associated with better performance in (instrumental) activities of daily living among older adults [45, 46]. Consequently, improvements in leg extensor power alone may not be sufficient to produce substantial gains in functional performance. Finally, these results should not be extrapolated to daily life competence: functional tests capture physical capacities under controlled conditions, and improvements in test performance do not necessarily translate to better real-world functioning [47].

4.5 | Limitations

In addition to the previously described limitations, including the initial improvements associated with exposure to a novel training stimulus and the absence of a non-exercising control group, several other factors should be considered when interpreting the present findings.

First, since only community-dwelling, physically and cognitively well-functioning older adults were included in our study, the generalizability of our findings to the broader older population is limited. In addition, since the intervention did not include sufficient force-deficient participants, no conclusions can be drawn regarding the effectiveness of F-V-based load prescription in older adults classified as force-deficient.

Second, F-V profiling and outcome assessments were performed only on the dominant leg, whereas both limbs completed the unilateral training program. Although this approach was chosen to reduce participant burden and testing duration and was supported by previous evidence indicating minimal inter-limb asymmetries in older adults, potential side-to-side differences in baseline characteristics and training-induced adaptations were not assessed. Therefore, asymmetrical adaptations cannot be entirely excluded and may have influenced performance in bilateral functional tasks such as the sit-to-stand and stair ascent tests.

Finally, although our velocity-based training protocol resulted in similar values for the calculated training volume indices between groups, technical limitations prevented us from assessing potential differences in mechanical work. While existing evidence indicates comparable adaptations following volume- and load-matched low- and high-load resistance training [48], calculating work as volume × load may underestimate actual mechanical work at light loads (i.e., the IND group in our study) [49]. Access to force and displacement data would have provided a more accurate representation of the mechanical demands and training stimulus between groups.

4.6 | Practical Applications

Our findings indicate that, among healthy, community-dwelling, high-functioning older adults classified as velocity-deficient according to our operational definition, training at a load targeting the deficit did not enhance maximal muscle power or functional performance more than training at a load deliberately not matched to the F-V deficit. From a practical standpoint, training load may be better guided by an older adult's preferences, abilities, and needs rather than by their F-V characteristics. Low-load, high-velocity training remains a valuable option, particularly when equipment is limited [50], or when working with exercise-naive or mobility-restricted older adults [51]. The observed variability in training preferences within our sample further highlights that personal preference may be the more meaningful driver of effective training prescription.

5 | Perspective

To our knowledge, our study is the first to classify force and velocity deficits in older adults relative to a sex-specific young reference group using a z-score approach. Further research is needed on the identification and classification of force and velocity deficits in aging populations. Establishing consensus on the most appropriate classification methodology represents an important next step. However, determining the optimal approach will likely require longitudinal research evaluating whether classifications based on young-reference benchmarking, within-sample distributions, or slope-based methods differ in their ability to predict functional outcomes and guide effective training prescription.

Future research should also investigate longer intervention durations to evaluate the effects of F-V profile-based training beyond the initial 12-week adaptation period. It remains unclear

whether F-V-based training should be implemented from the onset of an exercise program or after an initial preparatory phase, particularly in less-trained individuals. This question is relevant because substantial improvements typically occur during the early stages of a new training regimen, whereas gains tend to diminish over time. Consequently, exercise programming may play a relatively minor role in immediate adaptations to unfamiliar exercise but become increasingly important as training progresses.

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Ethics Statement

The study was approved by the Ethical Committee UZ/KU Leuven (S68434).

Consent

All participants provided written informed consent in accordance with the Declaration of Helsinki.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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